

Team Gold: Rogue-Like Algebra

Lab 2: Requirements & Specifications

Jillian Cequeria

CS 411W

Professor Kennedy

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1 Introduction

1.1 Purpose

The purpose of this document is to outline the requirements and specifications of *Rogue-Like Algebra* which is being developed. It is meant to be read and utilized by the developers and testers of the product.

1.2 Scope

Many students struggle to understand algebra due to traditional teaching methods that fail to engage learners effectively. In the 2020-21 school year, only 74% of over 2 million high schoolers passed their Algebra I class (U.S. Department of Education). This failure leads to frustration, disengagement, and a lack of confidence in math skills. They might attempt to seek out learning tools on their own, only to find that these traditional forms of learning still persist even through modern resources. *Figure 1* displays this dilemma, in which students seeking learning materials may find themselves in an unrewarding loop of learning and still require more assistance when these materials fail to help them.

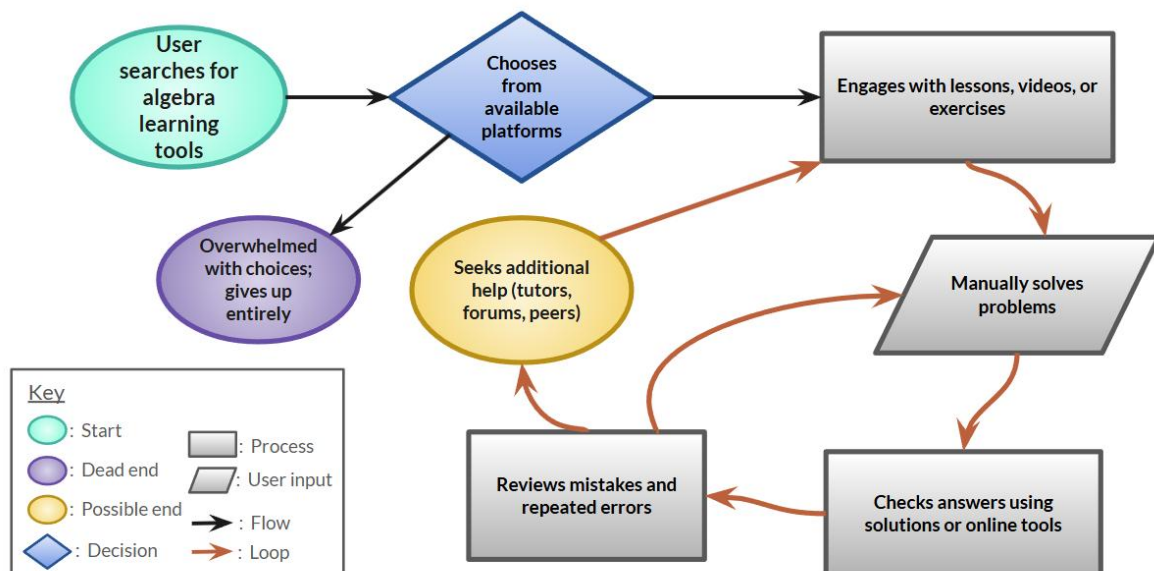


Figure 1: Current Process Flow

A solution must address the need for interactive, engaging, and adaptive learning tools that make algebra approachable and rewarding. The aim is to develop a game that will help those who struggle to understand algebra. *Rogue-Like Algebra* will be a turn-based, rogue-like adventure designed to gamify algebra. With dungeons, bosses, and power-ups all throughout,

the player will be as engaged in the world as they are with the math. By integrating strategic gameplay with problem-solving, learning can be transformed into an immersive and fun experience. The game aims to foster confidence in math skills while making algebra fun and engaging for middle and high school students.

1.3 Definitions

- **Adaptive Learning** – An educational method that adjusts content difficulty based on the learner’s performance to personalize the learning experience.
- **Algebraic Concepts** – Foundational mathematical principles involving variables, equations, expressions, and functions typically taught in middle and high school.
- **Continuous Integration (CI)** – A development practice where developers frequently merge code changes into a shared repository, triggering automated builds and tests to detect issues early.
- **Continuous Deployment (CD)** – A software release process that automatically delivers validated code changes to production, minimizing manual intervention and ensuring faster updates.
- **GScript** – A Python-like scripting language used in the Godot game engine, designed for creating game logic.
- **Git** – A distributed version control system that tracks changes in source code during software development, enabling collaboration and history tracking.
- **GitHub** – A cloud-based platform for hosting Git repositories. It offers tools for collaboration, issue tracking, code review, and CI/CD integration.
- **Godot** – A free and open-source game engine used for developing 2D and 3D games, known for its flexibility and ease of use.
- **IDE (Integrated Development Environment)** – A software suite that provides tools like a code editor, debugger, and compiler to streamline software development (e.g., VS Code).
- **Procedural Generation** – A method of creating data algorithmically rather than manually, often used for generating levels, puzzles, or content dynamically in games.
- **Rogue-like** – A subgenre of games typically featuring procedural generation, turn-based movement, and permanent death, emphasizing strategy and replayability.
- **Rust** – A systems programming language known for performance, safety, and concurrency.
- **Turn-Based Combat** – A gameplay mechanic where players and opponents take alternating turns to perform actions such as attacking or defending.

- **VS Code (Visual Studio Code)** – A popular, lightweight code editor developed by Microsoft, used for writing and debugging code.

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1.5 Overview

The rest of this document will explain the product’s purpose, functionality, user base, constraints, assumptions, and dependencies in a numbered format.

2 Overall Description

2.1 Product Perspective

Rogue-Like Algebra is a turn-based adventure game that transforms the challenge of learning algebra into an engaging and immersive experience, as shown in *Figure 2*. Designed to address the struggles many face with algebraic concepts, the game blends strategic dungeon-crawling with problem-solving, where players use algebra to attack, defend, and progress. Mistakes carry consequences—such as reduced combat efficiency or character penalties. The game's procedurally generated dungeons and adaptive difficulty ensure replayability and personalized challenge. Aimed primarily at middle and high school students, but accessible to all learners, *Rogue-Like Algebra* provides an educational alternative to traditional rote memorization by offering a gamified system that rewards creativity, risk-taking, and improvement. The game is available on the web and can be played in any modern browser. The project’s goal is to foster confidence in math skills while making algebra fun, approachable, and rewarding.

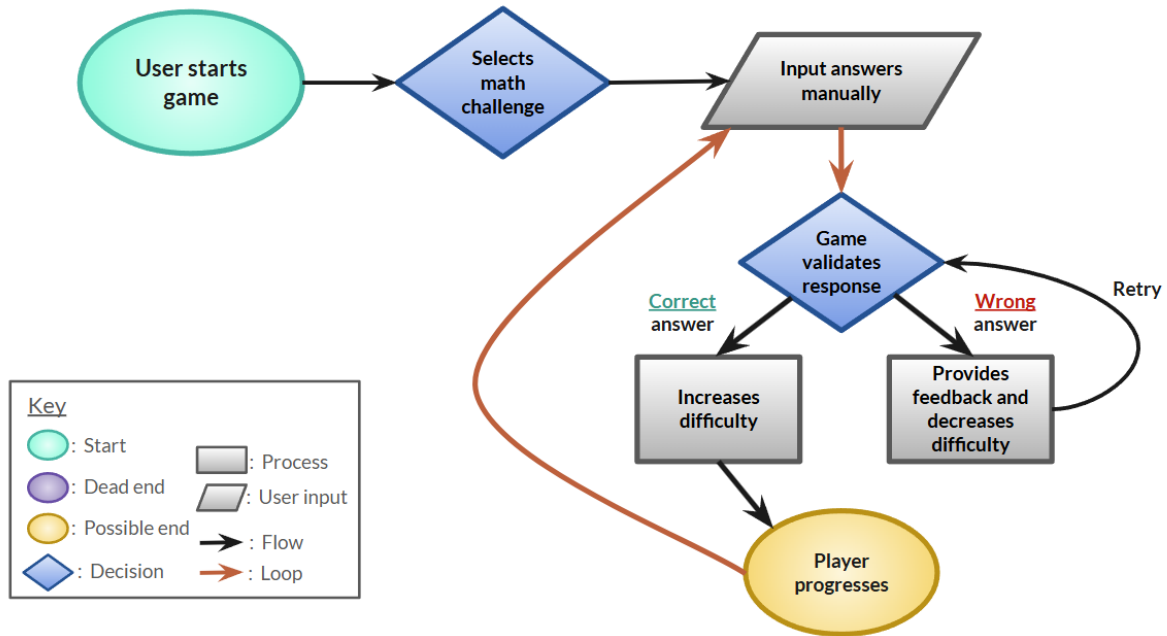


Figure 2: Solution Process Flow

2.2 Product Functions

The software will have major functionalities such as a character controller, enemy AI, pathfinding, procedurally generated problems, web integration, and an inventory system with items that may be equipped and upgraded.

2.3 User Characteristics

Rogue-Like Algebra is primarily for middle and high school students who will likely take an algebra course or an otherwise related math. STEM-oriented players and puzzle game enthusiasts will find the game tailored to their interests as well, making it both educational for new learners and entertaining for those already familiar with algebra. Though these are the groups of people most catered to for the game, other kinds of players can gain just as much value from it. Educators, the parents of learning students, casual and indie genre fans, and those who enjoy logic and strategy-based games can also benefit from the purpose of *Rogue-Like Algebra*.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

N/A